

CLOUD COMPUTING

Unit III: Cloud Based Services

Web Services, Cloud Providers & Cloud Infrastructure



SAGE
UNIVERSITY
—BHOPAL—

Unit III – Topics Covered

- What are Web Services? & Various Cloud Computing Services
- Cloud Service Providers
- Cloud Computing Demos – AWS, Microsoft Azure, Google Cloud
- Cloud Economics and Billing Models
- Cloud Infrastructure – Regions & Availability Zones
- Edge Locations & Regional Edge Cache



What are Web Services?

- A standardized way of enabling communication between two software applications over a network, typically the internet
- Uses standard protocols such as HTTP, XML, SOAP, or REST/JSON for exchanging data
- Allows applications built in different languages, on different platforms, to communicate with one another
- Cloud providers expose their infrastructure and platform capabilities as web services (APIs)
- Forms the interface through which users, applications, and management tools interact with cloud resources



Services Provided by Cloud Computing

- **Compute Services:** virtual servers, containers, and serverless functions for running applications
- **Storage Services:** object, block, and file storage for data of all types
- **Database Services:** managed relational and NoSQL databases
- **Networking Services:** virtual networks, load balancers, content delivery, and DNS
- **Security, Identity & Management:** access control, monitoring, and governance tools



Cloud Service Providers

- **Amazon Web Services (AWS):** the market leader, offering the broadest and deepest set of services
- **Microsoft Azure:** strong hybrid cloud and enterprise integration with Microsoft products
- **Google Cloud Platform (GCP):** known for data analytics, machine learning, and Kubernetes
- **IBM Cloud, Oracle Cloud, Alibaba Cloud:** notable enterprise and regional providers
- **Competition:** providers compete on pricing, global reach, service breadth, and specialized capabilities



Cloud Computing Demo – AWS

- **Amazon EC2:** launch and configure scalable virtual server instances
- **Amazon S3:** create buckets for scalable object storage
- **Amazon RDS:** provision a managed relational database
- **AWS Lambda:** run serverless functions triggered by events
- **Amazon VPC:** configure an isolated virtual network for AWS resources



Cloud Computing Demo – Microsoft Azure

- **Azure Virtual Machines:** deploy Windows or Linux virtual servers on demand
- **Azure Blob Storage:** store unstructured data such as documents, images, and backups
- **Azure SQL Database:** provision a fully managed relational database
- **Azure Functions:** build serverless, event-driven applications
- **Azure Virtual Network:** connect and secure Azure resources



Cloud Computing Demo – Google Cloud

- **Compute Engine:** create and manage scalable virtual machines
- **Cloud Storage:** store and retrieve data with a simple, unified object storage service
- **BigQuery:** run fast, serverless SQL queries on large datasets
- **Cloud Functions:** build lightweight, event-driven serverless applications
- **Virtual Private Cloud (VPC):** configure a private, isolated Google Cloud network



Cloud Economics

CapEx (Traditional IT)

Large upfront investment in hardware and data centers. Costs are fixed regardless of usage, and capacity must be over-provisioned for peak demand.

OpEx (Cloud Computing)

Pay only for resources consumed, with no upfront investment. Costs scale directly with usage, and capacity adjusts automatically to real demand.



Cloud Billing Models

- **Pay-As-You-Go:** pay only for resources consumed, billed hourly or per second, with no long-term commitment
- **Reserved Instances:** commit to a 1 or 3 year term in exchange for significantly discounted rates
- **Spot / Preemptible Instances:** bid on spare provider capacity at steep discounts, with the risk of interruption
- **Free Tier:** limited resources offered at no cost, useful for learning and small workloads
- **Basis:** billing is typically based on compute hours, storage used, and data transferred



Cloud Infrastructure Overview

Region

A geographic area containing multiple, isolated data centers, each grouped into Availability Zones.

Availability Zone

One or more discrete data centers with independent power, cooling, and networking within a Region.

Edge Location

A smaller site used to cache content closer to end users, reducing latency for delivery.

Regional Edge Cache

A larger cache layer between edge locations and the origin, holding less frequently accessed content.



Regions

- A Region is a geographic area – e.g. Mumbai, N. Virginia, or Singapore – containing multiple isolated data centers
- Each Region operates completely independently, for fault tolerance and stability
- Organizations choose Regions based on latency, cost, and data residency or compliance requirements
- Not all cloud services are available in every Region
- Major providers operate dozens of Regions across the globe



Availability Zones

- An Availability Zone (AZ) consists of one or more discrete data centers with independent power, cooling, and networking
- Each Region typically contains multiple AZs, physically separated from one another
- Distributing applications across multiple AZs protects against single data-center failures
- AZs within a Region are connected via high-speed, low-latency private links
- Designing for multi-AZ deployment is a core best practice for high availability



Edge Locations & Regional Edge Cache

Edge Location

A smaller site, located in major cities worldwide, used by content delivery networks (CDNs) to cache content closer to end users and reduce latency.

Regional Edge Cache

A larger cache layer positioned between edge locations and the origin server, holding content that is less frequently accessed but still needs fast delivery.



Unit III – Summary

- Web services provide the standardized interface through which cloud resources are accessed and managed
- Cloud computing offers compute, storage, database, networking, and security services
- AWS, Azure, and GCP are the leading providers, each offering comparable core services
- Cloud economics shifts IT spending from CapEx to OpEx, with flexible billing models
- Cloud infrastructure is organized into Regions, Availability Zones, Edge Locations, and Regional Edge Caches
- **Next:** Unit IV – Compute and Storage Services

